

$\Rightarrow$ Gamma function:-

$\Rightarrow$ if integer :- the

$$t^n = \frac{n!}{s^{n+1}}$$

but if in the fraction form then?

$$\sqrt{n+1} \rightarrow t^{-\frac{1}{2}}$$

$$= \frac{\sqrt{\frac{1}{2}+1}}{s^{n+1}} = \frac{\sqrt{\pi}}{s^{n+1}} - \frac{\sqrt{\pi}}{s^{\frac{1}{2}}} \cdot \sqrt{\frac{\pi}{s}}$$

Ans

$\rightarrow$ For positive numbers $n\sqrt{n}$

$$\frac{\sqrt{\frac{1}{2}+1}}{s^{n+1}}$$

$$\frac{1\sqrt{\frac{1}{2}}}{\sqrt{s^{\frac{3}{2}}}}$$

Ans: